The Isotropic Orthant Constant Has Order $n^{-1/2}$

A full asymptotic answer to Remark 3 of arXiv:1604.05351

Automated mathematics run `fa_banach_001`, lane 14

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Status. Candidate full solution, likely valid, pending human review. The proof resolves the conjectured asymptotic order; it does not determine the exact numerical value of the best constant.

1 Source question

Fradelizi, Meyer, and Yaskin [1] define in Remark 3 the best constant $\alpha_n > 0$ such that, for every isotropic convex body $K \subset \mathbb{R}^n$ and every orthonormal basis $u_1, \ldots, u_n$,

$$\left(\frac{|K \cap \{x : \langle x, u_i\rangle \geq 0,\ 1 \leq i \leq n\}|}{|K|}\right)^{1/n} \geq \frac{\alpha_n}{2}. \tag{1}$$

They prove $\alpha_n \geq n^{-1}$ and ask for a better estimate, writing that calculations suggest order $n^{-1/2}$. Their Hadamard-cube example gives the matching upper estimate for powers of two. The complete source passage is on p. 11 of the arXiv PDF:

the regular centered simplex Δ in $\mathbb{R}^n$ and a small nonempty cone C around the direction of a vertex of the simplex, then $|C \cap K| \sim n^n|(-C) \cap K|$, when the spherical measure of $C \cap S^{n-1}$ tends to 0.

3. It may be asked what is the best constant $\alpha_n > 0$ such that for any isotropic convex body K in $\mathbb{R}^n$ and any orthonormal basis $u_1, \ldots, u_n$, one has

$$\left(\frac{|K \cap \{x : \langle x, u_i\rangle \geq 0, 1 \leq i \leq n\}|}{|K|}\right)^{1/n} \geq \frac{\alpha_n}{2}.$$

It is clear and follows from Corollary 3 that $\alpha_n \geq n^{-1}$. Is there a better estimate? Some calculations on the regular simplex seem to indicate that α_n could be of the order of $n^{-\frac{1}{2}}$. Observe that if K is centrally symmetric and in John's position (instead of being isotropic) then the bound $n^{-\frac{1}{2}}$ trivially holds because then $B_2^n \subset K \subset \sqrt{n}B_2^n$ and the constant has the right order of magnitude as shown by the example of the cube:

Let $K = B_\infty^n = [-1, 1]^n$. If $n = 2^m$, it is well known that one can select n vertices $f_1, \ldots, f_n$ of some facet of K such that $(f_1, \ldots, f_n)$ are pairwise orthogonal. Let C be the convex cone generated by $f_1, \ldots, f_n$. Then

$$|K \cap C| = \frac{n^{n/2}}{n!},$$

and $\alpha_n \leq en^{-\frac{1}{2}}$.

Figure 1: Full-width crop of Remark 3 in the source paper.

We prove the suggested order in every dimension.

2 Proof intuition

The lower bound comes from an inball that is hidden in the isotropic normalization. Every one-dimensional marginal of the uniform measure on K is log-concave. Log-concavity prevents its variance from being generated by a very long, very light tail on only one side. Together with Grunbaum's halfspace inequality, this forces the supporting distance of K in every direction to be a fixed multiple of the common standard deviation. Hence K contains a centered Euclidean ball. Gaussian maximum entropy compares that standard deviation to $|K|^{1/n}$, and one sector of the ball lies in each orthant.

For the upper bound, the source already gives the correct cube construction when a Hadamard matrix of order n is available. We write an arbitrary n as a sum of distinct powers of two, place a Sylvester-Hadamard construction in each coordinate block, and take their product. A bounded binary entropy loss preserves the order $n^{-1/2}$.

3 Two elementary lower-bound lemmas

Lemma 1 (One-sided log-concave moment bound)**.** *Let $Y \geq 0$ have a log-concave probability density, and put $m = \mathbb{E}Y$. Then*

$$\mathbb{E}Y^2 \leq C_0 m^2, \qquad C_0 = 4 + \frac{8}{(\log 2)^2}. \tag{2}$$

Proof. The survival function $S(t) = \mathbb{P}(Y \geq t)$ is log-concave by the Prekopa-Leindler theorem [3]. Markov's inequality gives $S(2m) \leq 1/2$. Since $\log S$ is concave and $S(0) = 1$, for $t \geq 2m$,

$$S(t) \leq \exp\left(-\frac{(\log 2)t}{2m}\right).$$

Writing $a = (\log 2)/2$ and integrating the tail,

$$\begin{aligned}\mathbb{E}Y^2 &= 2\int_0^\infty tS(t)\,dt \\ &\leq 4m^2 + 2\int_0^\infty te^{-at/m}\,dt = \left(4 + \frac{2}{a^2}\right)m^2 = C_0 m^2.\end{aligned}$$

□

Lemma 2 (Centered inball at the covariance scale)**.** *Let X be uniform on a convex body $K \subset \mathbb{R}^n$ with centroid zero and covariance matrix $\sigma^2 I_n$. Then*

$$c_0\sigma B_2^n \subset K, \qquad c_0 = (1 + eC_0)^{-1/2}. \tag{3}$$

Proof. Fix a unit vector θ, put $Z = \langle X, \theta\rangle$, and let $b = h_K(\theta) = \sup_{x\in K}\langle x, \theta\rangle$. The marginal density of Z is log-concave by Prekopa-Leindler. Write

$$q = \mathbb{P}(Z \leq 0), \qquad M = \mathbb{E}Z_+ = \mathbb{E}(-Z)_+,$$

where the equality uses $\mathbb{E}Z = 0$. Grunbaum's centroid inequality gives $q \geq e^{-1}$ [2, 1]. The conditional random variable $Y = -Z \mid \{Z \leq 0\}$ has a log-concave density on $[0, \infty)$ and mean M/q. The preceding lemma therefore gives

$$\mathbb{E}[Z^2 \mathbf{1}_{\{Z\leq 0\}}] = q\mathbb{E}Y^2 \leq \frac{C_0 M^2}{q} \leq eC_0 b^2,$$

because $M = \mathbb{E}Z_+ \leq b$. Also

$$\mathbb{E}[Z^2 \mathbf{1}_{\{Z\geq 0\}}] \leq b\mathbb{E}Z_+ = bM \leq b^2.$$

Consequently $\sigma^2 = \mathbb{E}Z^2 \leq (1 + eC_0)b^2$, so $h_K(\theta) \geq c_0\sigma$. This holds for every $\theta \in S^{n-1}$, which is equivalent to (3). □

4 Main theorem

Theorem 1. *For every $n \geq 1$, the constant in (1) satisfies*

$$\frac{c}{\sqrt{n}} \leq \alpha_n \leq \frac{4e}{\sqrt{n}}, \qquad c = \frac{2}{\sqrt{2\pi e}\sqrt{1 + e(4 + 8/(\log 2)^2)}}. \tag{4}$$

In particular, $\alpha_n \asymp n^{-1/2}$, resolving the asymptotic-order question in Remark 3 of the source.

Lower bound. Let X be uniform on an isotropic convex body K. Thus $\mathbb{E}X = 0$ and $\mathrm{Cov}(X) = \sigma^2 I_n$. Nonnegativity of the relative entropy of the uniform density $|K|^{-1}\mathbf{1}_K$ with respect to the centered Gaussian of covariance $\sigma^2 I_n$ gives

$$0 \leq -\log|K| + \frac{n}{2}\log(2\pi\sigma^2) + \frac{n}{2},$$

and hence

$$\frac{\sigma}{|K|^{1/n}} \geq \frac{1}{\sqrt{2\pi e}}. \tag{5}$$

For the orthant $O = \{x : \langle x, u_i\rangle \geq 0, 1 \leq i \leq n\}$, the inball lemma yields

$$\frac{|K \cap O|}{|K|} \geq 2^{-n}|B_2^n|\frac{(c_0\sigma)^n}{|K|}.$$

The cube $[-n^{-1/2}, n^{-1/2}]^n$ lies in B_2^n, so $|B_2^n|^{1/n} \geq 2/\sqrt{n}$. Taking nth roots and using (5),

$$\left(\frac{|K \cap O|}{|K|}\right)^{1/n} \geq \frac{c_0}{\sqrt{2\pi e}\sqrt{n}}.$$

Comparison with (1) proves the lower bound in (4). $\square$

Upper bound. First let $d = 2^r$. Choose a Sylvester-Hadamard matrix H_d with first row equal to $(1, \dots, 1)$. Its columns $f_1, \dots, f_d$ are pairwise orthogonal vertices of the facet $\{x_1 = 1\}$ of $B_\infty^d = [-1, 1]^d$. For the orthonormal basis $u_i = f_i/\sqrt{d}$, its positive orthant is the cone $C_d = \mathrm{pos}(f_1, \dots, f_d)$. If $x = \sum_i a_i f_i$ with $a_i \geq 0$, then $x_1 = \sum_i a_i$; moreover $|x_j| \leq \sum_i a_i$ for every coordinate. Therefore

$$B_\infty^d \cap C_d = \mathrm{conv}(0, f_1, \dots, f_d).$$

Since $|\det H_d| = d^{d/2}$,

$$p_d := \frac{|B_\infty^d \cap C_d|}{|B_\infty^d|} = \frac{d^{d/2}}{2^d d!} \leq \left(\frac{e}{2\sqrt{d}}\right)^d. \tag{6}$$

For arbitrary n, write its binary expansion as $n = \sum_{d \in \mathcal{D}} d$, where the elements of $\mathcal{D}$ are distinct powers of two. Use the cube B_∞^n, split its coordinates into these blocks, and take the block-diagonal union of the Hadamard bases above. The orthant fraction is the product of the block fractions, so

$$p_n \leq \left(\frac{e}{2}\right)^n \exp\left(-\frac{1}{2}\sum_{d \in \mathcal{D}} d \log d\right). \tag{7}$$

If 2^m is the largest binary block, then $n < 2^{m+1}$ and

$$\begin{aligned}\sum_{d \in \mathcal{D}} d \log(n/d) &\leq (\log 2)\sum_{k=0}^{m} 2^k(m + 1 - k) \\ &< 4n \log 2.\end{aligned}$$

Thus $\sum d \log d > n \log n - 4n \log 2$. Substitution in (7) gives

$$p_n^{1/n} \le \frac{2e}{\sqrt{n}}.$$

The cube is isotropic, and α_n is twice the infimum of these nth-root orthant fractions. Hence $\alpha_n \le 4e/\sqrt{n}$. $\square$

5 Verification, novelty, and limitations

The script `code/verify_hadamard_blocks.py` verifies the exact Hadamard orthogonality and determinant identities for powers of two through 128, checks the factorial bound numerically, and tests the binary entropy and product inequalities for every $1 \le n \le 100{,}000$. These computations check finite-dimensional algebra but are not used to replace any analytic step of the proof.

The local run indexes and three bounded web-search rounds on 21 July 2026 used the exact source phrase, title, authors, arXiv id, and combinations of *isotropic convex body*, *positive orthant*, *orthant probability*, and the $n^{-n/2}$ scale. Only the source and unrelated isotropic/log-concave material were found. No later answer or matching proof was located. This is not an exhaustive citation search, so novelty confidence is moderate.

The result determines the sharp order of α_n, not its exact value or optimal leading constant. Human review should focus on the survival-function argument in Lemma 1, the inball implication in Lemma 2, and the binary-block entropy estimate in the upper bound.

References

[1] M. Fradelizi, M. Meyer, and V. Yaskin, *On the volume of sections of a convex body by cones*, Proc. Amer. Math. Soc. 145 (2017), 3153–3164; arXiv:1604.05351.

[2] B. Grunbaum, *Partitions of mass-distributions and of convex bodies by hyperplanes*, Pacific J. Math. 10 (1960), 1257–1261.

[3] A. Prekopa, *Logarithmic concave measures with applications to stochastic programming*, Acta Sci. Math. (Szeged) 32 (1971), 301–316.